1. Evaluation protocol

One-year-ahead performance was measured on three untouched rolling origins: December 2020, 2021 and 2022, predicting calendar years 2021, 2022 and 2023.

Chronological experiment

- Eight compact SARIMA specifications were compared only on earlier 2018-2020 forecasts.
- Each evaluation fold was refitted using labels fully observed by its forecast origin.
- The GRU used the last 24 eligible training samples for temporal early stopping.
- Primary metric: RMSE. Secondary metrics: MAE, sMAPE, bias and residual autocorrelation.
- Seasonal naive (same month one year earlier) was retained as a diagnostic benchmark.

Decision rule

The model with the lowest aggregate rolling-origin RMSE supplies the single predictions CSV. No averaging or blending is used.

2. Classical model: SARIMA

The classical baseline is a seasonal autoregressive integrated moving-average model fitted only to food-inflation history available at each origin. It produces a 12-step forecast without requiring unknown 2024 explanatory variables.

Setting	Value	Reason
Order	(1,1,1)	Non-seasonal AR, difference and MA
Seasonal order	(1,0,1,12)	Annual monthly seasonality
Selection	8 candidates; 2018-2020	Pre-evaluation rolling origins
Outputs	12-step forecast	Matches submission horizon
Final AIC	645.1	In-sample fit diagnostic only

ADF/KPSS stationarity evidence

At level, ADF p=0.220 does not reject a unit root, while KPSS p=0.078 does not reject stationarity at 5%, giving mixed evidence. After first differencing, ADF p=0.000002 and KPSS p=0.100 support stationarity; this justifies d=1.

Strengths and risk

SARIMA is reproducible, interpretable and appropriate for a short monthly series, but it is univariate and cannot learn nonlinear global-shock interactions. The GRU supplies that multivariate comparison.

3. Deep model: compact NumPy GRU

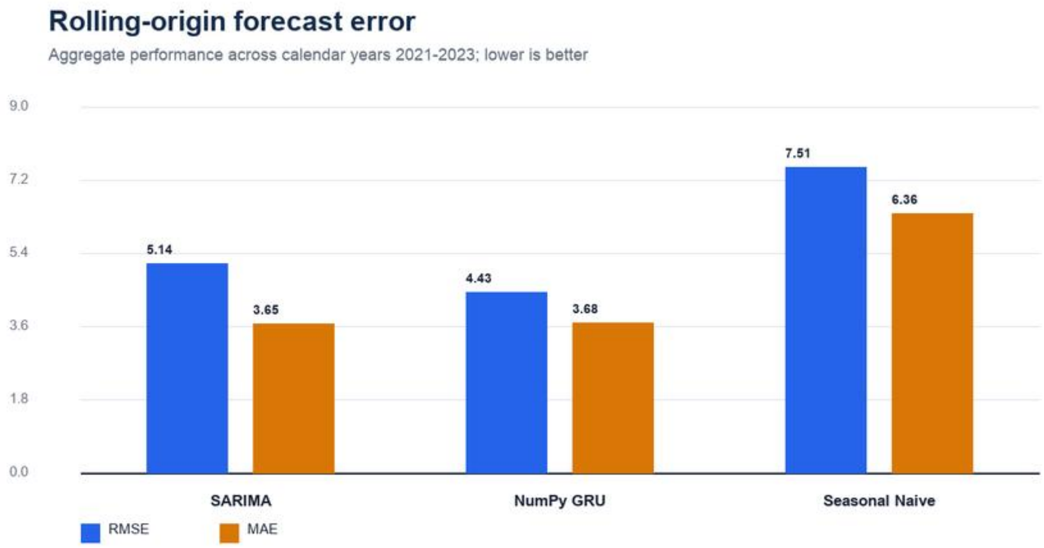
The deep model is a gated recurrent unit that consumes the latest 24 monthly observations and directly emits 12 forecasts. It is implemented in NumPy so every gate, gradient and training decision is auditable.

Hyperparameter	Value	Small-data control
Context	24 months	Two annual cycles
Channels	9	Compact economically selected inputs
Hidden units	16	Limits capacity
Parameters	1,452	Far below transformer scale
Input dropout	15%	Reduces feature co-adaptation
L2 penalty	0.0001	Weight shrinkage
Optimiser	Adam; gradient clip 5	Stable recurrent training
Stopping	patience 60	Temporal validation only

A direct output head was preferred to recursive prediction because recursive monthly errors compound across a full year.

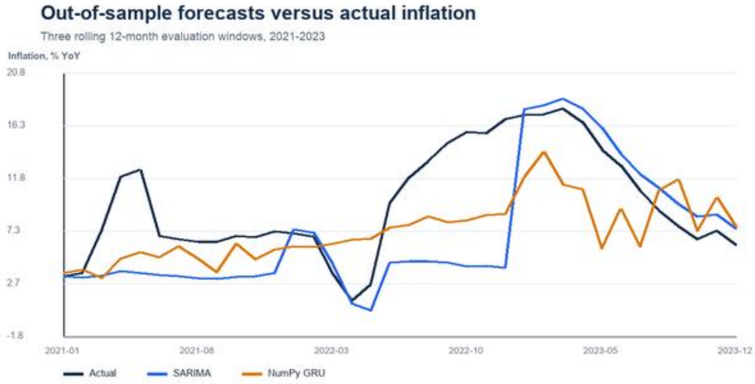
4. Rolling-origin results

Model	RMSE	MAE	sMAPE	Bias
SARIMA	5.141	3.654	50.76%	-2.670
NumPy GRU	4.434	3.682	42.99%	-2.397
Seasonal Naive	7.508	6.364	73.28%	-2.736

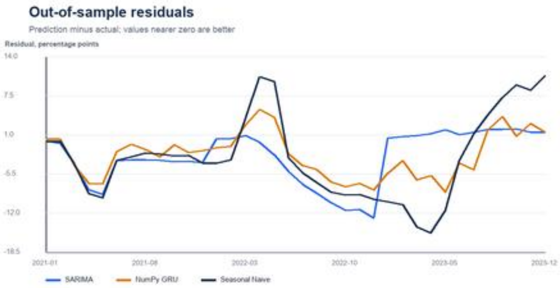


Winner: NumPy GRU. Its aggregate 2021-2023 RMSE is 4.434. The choice follows the pre-declared RMSE rule.

5. Diagnostics, final choice and limitations



Residual diagnostics



Model	Lag-1 ACF	Lag-12 ACF	Interpretation
SARIMA	0.793	-0.227	Lower absolute values indicate less remaining structure
NumPy GRU	0.739	-0.754	Lower absolute values indicate less remaining structure
Seasonal Naive	0.854	-0.818	Lower absolute values indicate less remaining structure

Honest conclusion

The NumPy GRU won. Its gated sequence representation captured nonlinear persistence that the univariate SARIMA missed, while dropout, L2 regularisation and early stopping controlled the small-sample risk.

Pitch spot-check

The June-October easing follows 2023's actual decline from 12.9% to 6.5%; the NumPy GRU extends that seasonal/base-effect disinflation, although its steeper 2024 decline from 13.4% to 3.7% is a key forecast uncertainty.

Limitations include only 276 target observations, possible structural breaks, and no known 2024 explanatory values. Results are forecasts, not causal estimates.